🧩 Task 1 – Analyze and Fix Incorrect Permissions

Create a file:

touch app.log
chmod 777 app.log
ls -l app.log

🎯 Requirement:

- Only owner should read & write.
- No one else should access it.

✔ Fix using **numeric method**
✔ Verify using `ls -l`

🧩 Task 2 – Directory Access Without Execute

Create:

mkdir secure_dir
chmod 644 secure_dir

🎯 Try:

- `ls secure_dir`
- `cd secure_dir`

❓ Observe:

Why does it fail?

✔ Fix the minimum required permission to:

- Allow entering directory

- Prevent file creation inside

🧩 Task 3 – Test File Deletion Rule

Create:

mkdir test_delete
chmod 777 test_delete
touch test_delete/sample.txt
chmod 000 test_delete/sample.txt

🎯 Try deleting file:

rm test_delete/sample.txt

❓ Does it delete?

Now:

chmod 555 test_delete
touch test_delete/new.txt

❓ Why does creation fail?

✔ Explain directory vs file permission behavior.

🧩 Task 4 – Ownership & Group Management

Create group and users (if not already created):

sudo groupadd devteam

Create file:

touch project.txt

🎯 Requirements:

- Owner → user1

- Group → devteam
- Only owner and group can read
- Only owner can write

✔ Use `chown` and `chmod`
✔ Verify

🧩 Task 5 – SGID on Shared Directory

Create shared project directory:

sudo mkdir /team_project
sudo chown root:devteam /team_project

🎯 Requirements:

- Only devteam members can access
- New files must inherit group automatically
- Others must have no access

✔ Apply correct **4-digit permission**
✔ Verify group inheritance

🧩 Task 6 – Sticky Bit Protection

Create shared temp directory:

mkdir shared_temp
chmod 777 shared_temp

🎯 Problem:

Any user can delete others' files.

✔ Apply sticky bit
✔ Verify permission string shows `t`

Test with two users:

- User1 creates file
- User2 attempts deletion

🧩 Task 7 – SUID Verification (Effective UID)

Create C program that prints:

- Real UID
- Effective UID

Compile it.

🎯 Requirements:

- Owner should be root
- Enable SUID
- Run as normal user

✔ Verify Effective UID changes
✔ Explain why Real UID does not change

🧩 Task 8 – SGID on Executable

Create program to print:

- Real GID

- Effective GID

🎯 Requirements:

- Change group to devteam
- Enable SGID
- Run as normal user

✔ Verify Effective GID changes

🧩 Task 9 – umask Behavior Testing

Check current umask:

umask

Set:

umask 0027

Create:

touch fileA
mkdir dirA

🎯 Tasks:

- Calculate expected permission manually
- Verify using `ls -l`
- Reset umask to default

🧩 Task 10 – Complete Real-World Secure Setup (Final Challenge)

🎯 Scenario:

You are a DevOps engineer.

Create directory `/dev_secure` with:

✔ Accessible only to devteam
✔ New files inherit group
✔ No outsider access
✔ No deletion of others' files
✔ Owner full access
✔ Group read & write

You must use:

- chown
- chmod
- SGID
- Sticky Bit
- 4-digit format

After setup:

- Test with two users
- Verify permission string
- Verify group inheritance
- Verify deletion restriction
